

An explicit formula for Campbell's variable-depth recurrence

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Abstract

We give an explicit formula for the nested recurrence proposed by John M. Campbell, in which the number of iterations is itself a preceding sequence term. Its even and odd subsequences consist of alternating linear pieces and plateaux on powers-of-three scales. The proof identifies every prescribed inner orbit after four steps: it is then fixed or two-periodic, and the iteration depth selects the required phase. We also obtain exact dilation and the lower and upper limits of the ratio of the sequence to its index.

1 The recurrence and its explicit formula

Campbell's proposed recurrence¹ is

$$b(1) = 1, \quad T_n(x) = n - b(x), \quad b(n) = T_n^{b(n-1)}(n-1) \quad (n \geq 2). \quad (1)$$

Here the exponent denotes iteration of the map $x \mapsto n - b(x)$ with n fixed. The first terms are

$$1, 1, 2, 3, 2, 3, 4, 5, 6, 7, 6, 9, 6, 9, 6, 9, 8, 9, 10, \dots$$

Theorem 1. *The recurrence defines a unique positive integer sequence. For $n \geq 2$, let s be the unique power of 3 in the applicable interval below. Then*

$$\begin{aligned} b(n) &= \min(n - s, 3s), & n \text{ even}, & \quad 2s \leq n < 6s, \\ b(n) &= \max(2s, n - 3s), & n \text{ odd}, & \quad 3s \leq n < 9s. \end{aligned} \quad (2)$$

For every $n \geq 2$, the orbit starting at $n - 1$ satisfies

$$T_n^6(n-1) = T_n^4(n-1). \quad (3)$$

Its preperiod is at most four and its eventual period is one or two. The bound four is attained.

For example, at scale $s = 9$ the even subsequence follows $n - 9$ from $n = 18$ through 36, and then stays at 27 through 52. The odd subsequence stays at 18 from 27 through 45, and then follows $n - 27$ from 47 through 79. Their interleaving explains the clusters.

Well-definedness. Suppose the prefix through $n - 1$ is defined with $1 \leq b(k) \leq k$. For $1 \leq x \leq n - 1$ we have $1 \leq n - b(x) \leq n - 1$. Thus T_n maps the known index set into itself, and any positive finite iterate exists uniquely. Strong induction proves well-definedness and $1 \leq b(n) \leq n - 1$ for $n \geq 2$. In fact, for $n \geq 3$ the map takes values in $[2, n - 1]$: at $x = 1$ its value is $n - 1$, while at $x \geq 2$ the stronger bound $b(x) \leq x - 1$ gives $n - b(x) \geq n - x + 1 \geq 2$.

2 The candidate function and short orbits

Define $F(1) = 1$ and define $F(n)$ by the right side of (2) for $n \geq 2$. Its scale is unique because the respective half-open intervals partition the eligible even and odd indices. Directly,

$$1 \leq F(n) \leq n - 1, \quad F(n) \equiv n + 1 \pmod{2} \quad (n \geq 2). \quad (4)$$

The following four rules hold on closed intervals; the formulas agree at every shared endpoint. In each row t is a power of 3.

¹John M. Campbell, [MetaMetaV32.pdf](#), unpublished note communicated 29 September 2026.

Parity of x	Interval	$F(x)$
even	$2t \leq x \leq 4t$	$x - t$
even	$4t \leq x \leq 6t$	$3t$
odd	$3t \leq x \leq 5t$	$2t$
odd	$5t \leq x \leq 9t$	$x - 3t$

At $x = 6t$, for instance, the next even scale $3t$ gives the same value $3t$; at $x = 9t$ the next odd scale gives $6t$.

Fix n and put $U_n(x) = n - F(x)$ on $1 \leq x \leq n - 1$. By (4), this interval is invariant. Write $x_0 = n - 1$ and $x_{j+1} = U_n(x_j)$.

Lemma 1. *Let $s \geq 3$ be a power of 3 in the interval associated with n . The orbit is given by the tables below. In particular, $x_6 = x_4$; if n is even then $x_4 = F(n)$, and if n is odd then $x_5 = F(n)$.*

In the first table set

$$A = \min(n - 2s/3, 5s/3 + 1), \quad B = \min(n - 2s/3, 2s).$$

Rows are closed for convenience, and are intersected with $2s \leq n < 6s$ or $3s \leq n < 9s$, respectively. Adjacent rows agree at shared endpoints.

Even n							
	Interval for n	x_1	x_2	x_3	x_4	x_5	x_6
E1	$[2s, 3s + 1]$	$s + 1$	$n - 2s/3 - 1$	A	$n - s$	B	$n - s$
E2	$[3s + 1, 10s/3]$	$n - 2s$	$7s/3$	$n - 4s/3$	$n - s$	$2s$	$n - s$
E3	$[10s/3, 4s]$	$n - 2s$	$n - s$	$2s$	$n - s$	$2s$	$n - s$
E4	$[4s, 5s + 1]$	$n - 2s$	$3s$	$n - 2s$	$3s$	$n - 2s$	$3s$
E5	$[5s + 1, 6s]$	$3s + 1$	$n - 2s - 1$	$n - 2s$	$3s$	$n - 2s$	$3s$
Odd n							
	Interval for n	x_1	x_2	x_3	x_4	x_5	x_6
O1	$[3s, 4s + 1]$	$s + 1$	$n - 2s/3 - 1$	$5s/3 + 1$	$n - s$	$2s$	$n - s$
O2	$[4s + 1, 13s/3]$	$n - 3s$	$10s/3$	$n - 7s/3$	$n - s$	$2s$	$n - s$
O3	$[13s/3, 5s]$	$n - 3s$	$n - s$	$2s$	$n - s$	$2s$	$n - s$
O4	$[5s, 6s + 1]$	$n - 3s$	$4s$	$n - 3s$	$4s$	$n - 3s$	$4s$
O5	$[6s + 1, 7s]$	$3s + 1$	$n - 2s - 1$	$n - 3s$	$4s$	$n - 3s$	$4s$
O6	$[7s, 8s + 1]$	$3s + 1$	$n - 2s - 1$	$n - 3s$	$n - 3s$	$n - 3s$	$n - 3s$
O7	$[8s + 1, 9s]$	$3s + 1$	$n - 2s - 1$	$5s + 1$	$n - 3s$	$n - 3s$	$n - 3s$

Proof. Use the four branch rules with $t = s/3, s, 3s$, all powers of 3. The parities alternate for even n , starting with odd x_0 ; for odd n all indices x_j are even. All listed arguments remain in $[1, n - 1]$.

We detail E1, which involves the two minima. Here $5s/3 \leq 2s - 1 \leq n - 1 \leq 3s$, so the odd linear branch at scale $s/3$ gives $F(n - 1) = n - s - 1$ and $x_1 = s + 1$. Since $2s/3 \leq s + 1 \leq 4s/3$, we get $F(x_1) = 2s/3 + 1$ and $x_2 = n - 2s/3 - 1$. This odd index lies in $[s, 3s]$, and hence

$$F(x_2) = \max(2s/3, n - 5s/3 - 1), \quad x_3 = A.$$

Both arguments defining A are at least $4s/3$, and $A \leq 5s/3 + 1 \leq 2s$. The even plateau at scale $s/3$ therefore gives $F(x_3) = s$ and $x_4 = n - s$. This odd index lies in $[s, 3s]$, so $F(x_4) = \max(2s/3, n - 2s)$ and $x_5 = B$. Finally, $4s/3 \leq B \leq 2s$ gives $F(x_5) = s$ and $x_6 = n - s$.

The remaining rows follow by the same substitutions. To illustrate the change of scale in O7, $x_1 = 3s + 1$ gives $F(x_1) = 2s + 1$, so $x_2 = n - 2s - 1 \in [6s, 7s - 1]$. The even linear branch at scale $3s$ gives $F(x_2) = n - 5s - 1$ and $x_3 = 5s + 1$. The even plateau at scale s then gives $x_4 = n - 3s$; this lies in $[5s + 1, 6s]$ and is fixed by U_n . Each of the other rows uses the displayed branch intervals directly. $\square$

3 Proof of the formula

We prove $b(n) = F(n)$ by strong induction. Direct evaluation through $n = 8$ gives $(1, 1, 2, 3, 2, 3, 4, 5)$, as required. For $n \geq 9$, assume equality at every smaller index. Then $T_n = U_n$ throughout $[1, n - 1]$. Since this interval is invariant, their entire starting orbits agree.

Let $d = b(n - 1) = F(n - 1)$. Equation (4) gives $d \equiv n \pmod{2}$. For even $n \geq 10$, the table gives $d \geq 4$: in E1, $d = n - s - 1 \geq n/2 - 1 \geq 4$; in E2–E4, $d = 2s \geq 6$; in E5, $d = n - 3s - 1 \geq 2s \geq 6$. For odd n , it gives $d \geq 5$: in O1, $d = n - s - 1 \geq 2s - 1 \geq 5$; in O2–O4, $d = 3s \geq 9$; in O5–O7, $d = n - 3s - 1 \geq 3s \geq 9$.

Since $x_6 = x_4$, determinism implies $x_{j+2} = x_j$ for every $j \geq 4$. For even n , d is even and at least four, so $b(n) = x_d = x_4 = F(n)$. For odd n , d is odd and at least five, so $b(n) = x_d = x_5 = F(n)$. This closes the induction.

The same lemma proves (3) for $n \geq 9$. For $n = 2, 3, 4, 5, 6, 7, 8$, direct calculation gives $x_4 = x_6$ equal to $1, 2, 3, 4, 3, 4, 5$, respectively. The bound four is attained at $n = 24$:

$$23 \mapsto 10 \mapsto 17 \mapsto 16 \mapsto 15 \mapsto 18 \mapsto 15 \mapsto 18 \mapsto \dots$$

The first six displayed values are distinct. This finishes the proof of Theorem 1. $\square$

4 Scaling, fluctuations and fixed points

Corollary 1. *For every $n \geq 2$, $b(3n) = 3b(n)$. Moreover,*

$$\liminf_{n \rightarrow \infty} \frac{b(n)}{n} = \frac{2}{5}, \quad \limsup_{n \rightarrow \infty} \frac{b(n)}{n} = \frac{3}{4}.$$

Proof. Multiplying n by 3 preserves parity, sends its scale s to $3s$, and multiplies either expression in (2) by 3. On even indices, $b(n)/n \in [1/2, 3/4]$; on odd indices it lies in $[2/5, 2/3]$. The values $n = 5 \cdot 3^k$ attain $2/5$, and $n = 4 \cdot 3^k$ attain $3/4$, proving the two limits. $\square$

For even $n \geq 4$, the eventual period is exactly two, since the two phases have opposite parity; at $n = 2$ the orbit is fixed. For odd n , the tables give

$$x_4 = \begin{cases} n - s, & 3s \leq n \leq 5s, \\ 4s, & 5s \leq n \leq 7s, \\ n - 3s, & 7s \leq n < 9s, \end{cases} \quad x_5 = \max(2s, n - 3s).$$

Consequently the prescribed orbit is eventually fixed exactly when $n = 3s$ or $7s \leq n < 9s$. At the other odd indices its period is two. The small scale $s = 1$ agrees by direct calculation at $n = 3, 5, 7$.

5 Computational support

The accompanying package provides two complementary checks. The program `explore.py` generates the original recurrence through 10^6 terms without using the closed formula, compares every term with it, and separately performs literal nesting through 5000 terms. All comparisons agree. The program `check_proof.py` checks the twelve orbit rows using exact rational Fourier–Motzkin elimination, including the strict inequalities needed to negate an implication. It verifies all 72 transitions over their full parameter regions, together with interval coverage, index bounds, parity, selected outputs, depth bounds and the small cases.

These computations supplement the written induction proof. This result has not yet been formalized in Lean. AI assistance was used for exploration, proof development, code generation and writing; the package records the arithmetic checks performed on the resulting argument.

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